DS 4420: Machine Learning and Data Mining 2 **(Due: Apr. 6 2026)**

Homework Assignment #6

Professor: Eric Gerber *Name:* ______________________

Instructions: Please include the following information on the first page of your completed homework write-up:

1. Your name
2. DS 4420
3. Homework #6

You will submit **up to four files** to Gradescope for this homework:

- Handwritten/latex typeset answers may be included in their own **.pdf** file or as part of either of the below
- A **.ipynb** file with all python code/output
- A **.pdf** file knitted from an included **.rmd** with all R code/output

Answers that are not supported by reasoning/work will not receive full credit. **Homework is due by 11:59 pm, via Gradescope, on the date above.** Late submissions will **not** be accepted, but you may receive extra credit for early submission (see syllabus for details).

You will also be graded on organization/neatness of the submitted files.

Multivariate Normal (10 points)

(1) For this problem you will make use of the **ds4420_spotify.csv** file on Canvas. Assume we are interested in three features: artist popularity (y_1), track popularity (y_2), and song duration (y_3). We will further assume that these three features follow a multivariate Gaussian distribution:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \sim MVN_3(\mu, \Sigma)$$

Complete all of the following tasks in **either Python or R (your choice)**.

(a) Compute the MLEs for μ and Σ from the data, save them and print them out. You will treat these as the true values for the purposes of this problem.

(b) Taylor Swift is perhaps the most popular artist ($y_1 = 100$) on Spotify and is planning to make a song 3 minutes long ($y_3 = 180$). Find the conditional distribution of her new song's track popularity, $p(y_2|y_1, y_3)$, and print out the conditional mean and variance.

(c) Plot out the conditional distribution for her new song's track popularity. What is the likely range for how popular her new song will be?

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Basic Sampling (30 points)

(2) As in the example in class, suppose you are interested in θ, the probability your crush is interested in you. You text your crush. We have established that it makes sense that:

$$y|\theta \sim Bern(\theta)$$
$$p(y|\theta) = \theta^y(1-\theta)^{1-y}$$

(a) To make our lives more difficult (maybe we're masochists), instead of the convenient beta distribution for θ, we are going to assume a priori that θ follows the logit-normal distribution:

$$\theta \sim LN(\mu, \sigma)$$
$$p(\theta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\left(\log\left(\frac{\theta}{1-\theta}\right)-\mu\right)^2}{2\sigma^2}} \frac{1}{\theta(1-\theta)}, \qquad 0 < \theta < 1$$

This distribution has no closed form and cannot be sampled from directly. In **R**, use **rejection sampling** to sample 10000 values from this prior distribution with $\mu = 1$ and $\sigma = 2$.

Rejection Sampling Algorithm:

- Propose samples v from $q(v) \sim Beta(1, 1)$ (equivalent to $Unif(0, 1)$)
- Accept each proposed sample with probability $\frac{p(v)}{kq(v)}$, where $k = 5$
- Repeat until you have 10000 accepted samples

Verification:

- Before implementing, verify that $kq(v) \geq p(v)$ for all $0 < v < 1$ by plotting both functions on the same axes
  - **NOTE:** Be careful with numerical stability: keep your θ values strictly between 0 and 1 (e.g., use `seq(0.001, 0.999, ...)`).
- After sampling, verify that your samples follow $p(\theta)$ by overlaying a histogram of your samples with the theoretical density $p(\theta)$

(b) In a Bayesian context, we know that the posterior distribution for θ in this model must be proportional to the joint:

$$p(\theta|y) \propto p(y, \theta) = p(y|\theta)p(\theta) = \theta^y(1-\theta)^{1-y} \times \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\left(\log\left(\frac{\theta}{1-\theta}\right)-\mu\right)^2}{2\sigma^2}} \frac{1}{\theta(1-\theta)}$$

This is impossible to marginalize mathematically. In **R**, use **both rejection and ancestral sampling** to generate samples from the joint distribution. That is:

- Use rejection sampling to sample θ from $p(\theta)$. (You can use your samples from (a)).
- Given those samples, use the `rbinom()` function to sample from $y|\theta$
- Create two visualizations: (1) the theoretical joint distribution $p(y, \theta)$ computed on a grid of θ values for $y \in \{0, 1\}$, and (2) the empirical joint distribution from your ancestral samples as a heatmap/2D histogram and discuss if your sampler did a good job
- Comment on what the next step is; how will you use the samples from $p(y, \theta)$ to eventually find the posterior $p(\theta|y)$? **Hint**: If you observe $y = 0$, which samples from your joint distribution are relevant for the posterior?

(Extra Credit: 5 points) Do both parts (a) and (b) in Python.

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Metropolis Sampling (30 points)

(3) Suppose you wish to sample from a bivariate normal distribution:

$$y|\mu, \Sigma \sim MVN_2(\mu, \Sigma)$$

(a) Using the isotropic Guassian as the proposal distribution, write a function in **R** called `gaussian_metropolis()` to sample from this distribution using the Metropolis algorithm. I'll start it for you:

```
library(mvtnorm) # dmvnorm requires the mvtnorm package

p <- function(y, mu, Sigma) {
  dmvnorm(y, mean = mu, sigma = Sigma)
}

gaussian_metropolis <- function(n, mu, Sigma, sigma = 2, burn = 0.18) {
    # you can do the rest
```

The main arguments of the function are:

- n (the number of samples)
- μ (the mean vector of y)
- Σ (the covariance matrix of y)
- σ (the standard deviation of the proposal)

The function should return a matrix with n new samples from the target bivariate normal distribution. Make sure to keep track of the Metropolis acceptance ratio and to include an 18% burn-in.

Hint: you should be able to do this very quickly by editing some of the code from lecture.

(b) Use your `gaussian_metropolis()` function to sample 10000 samples from a bivariate normal distribution with mean $\mu = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$ and covariance $\Sigma = \begin{bmatrix} 1 & -.6 \\ -.6 & 1 \end{bmatrix}$. Fiddle with the proposal standard deviation σ until you have an acceptance ratio around 23.4% (or, at least between 20% and 50%).

Then verify the convergence of your chains with plots, assess if the chains are mixing well with the ACF plots, and calculate the mean and covariance matrix of your samples to compare with the true parameters. Do you trust that your samples came from the target distribution? Explain why or why not.

(Extra Credit: 5 points) Do both parts (a) and (b) in Python.

Note: Identifying Conditional Posteriors

Imagine your data likelihood is the $y|\mu$ defined in the previous problem with $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$.

Given an **improper** uniform prior on μ, the joint posterior of $\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$ is:

$$\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} |y \sim MVN_2 \left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \right)$$

You can prove (using the notes from Lecture 8), that the conditional posteriors for the means are:

$$\mu_1|\mu_2, y \sim N(y_1 + \rho(\mu_2 - y_2), \sqrt{1-\rho^2})$$
$$\mu_2|\mu_1, y \sim N(y_2 + \rho(\mu_1 - y_1), \sqrt{1-\rho^2})$$

Gibbs Sampling (30 points)

(4) Assume you measure your first data point, $y = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

(a) In **R**, write a function to use Gibbs Sampling to sample 10000 samples from the joint posterior distribution of $p(\mu_1, \mu_2|y)$. Keep $\rho = -.6$ and use the function `rnorm()` to iteratively sample the conditional posteriors defined in the above **"Identifying Conditional Posteriors"** section.

Then verify the convergence of your chains with plots and assess if the chains are mixing well with the ACF plots.

(b) Note that you know what the true joint posterior mean and covariance matrix are. Calculate the mean and covariance matrix of your samples and compare to the true parameters. Do the mean and covariance matrix of your Gibbs samples match the truth? Discuss (in a sentence or two) the implications of defining an improper prior, especially the drawbacks and benefits, if any.

(Extra Credit: 5 points) Do part (a) in Python.